Diabetes Prediction Using Machine Learning (Logistic Regression)

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Abstract

Diabetes is one of the most common chronic diseases affecting people worldwide. This project develops a machine learning model to predict diabetes using patient health attributes. Logistic Regression, combined with median imputation and feature scaling, was used for classification.

Introduction

Machine learning can help identify patterns associated with diabetes. This project uses the PIMA/Kaggle diabetes dataset to classify patients as diabetic or non-diabetic.

Objectives

Study diabetes prediction, preprocess data, train a Logistic Regression model, evaluate performance, and identify key predictive factors.

Dataset Description

Features include Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age, and Outcome.

Methodology

Data collection, preprocessing with median imputation and standardization, model training using Logistic Regression, and evaluation using classification metrics.

Results

Accuracy: 0.734, Precision: 0.603, Recall: 0.704, F1-Score: 0.650, ROC-AUC: 0.813.

Key Findings

Glucose and BMI were among the most influential features. Data preprocessing improved model performance.

Limitations

Limited dataset size and not suitable as a replacement for clinical diagnosis.

Future Scope

Use Random Forest, XGBoost, larger datasets, and deploy a web application.

Conclusion

The model achieved satisfactory performance and demonstrated the usefulness of machine learning for early diabetes risk prediction.